

Ron Nyonje

Backend Engineer | Python • Django • Celery • Postgres • Docker • Kubernetes

📍 Nairobi, Kenya | 🌐 Open to Remote | ✉️ ron.nyonje@strathmore.edu |
🔗 [Portfolio](#) | 🔗 [LinkedIn](#) | 🔗 [GitHub](#)

Profile

Results-driven Backend Engineer with strong experience building and scaling web applications using Python, Django REST Framework, Celery, and PostgreSQL/MySQL. Skilled in writing clean, testable code (TDD), implementing scalable APIs, and deploying with Docker and Kubernetes in cloud environments. Hands-on experience with LLM frameworks (LangChain), CI/CD pipelines, and cross-functional collaboration. Strong Linux background, self-starter mindset, and a passion for solving real-world problems with technology.

Technical Skills

- Languages: Python 3, JavaScript (TypeScript, ReactJS)
- Backend: Django, Django REST Framework, Celery, Redis, ClickHouse
- Databases: PostgreSQL, MySQL
- DevOps: Docker, Docker Compose, Kubernetes, GitLab CI/CD, GCP
- AI/LLM: LangChain, LangGraph (exploratory projects)
- Testing/Quality: Pytest, unittest, TDD, Flake8, GitLab Boards (Scrum)
- Other: Pandas, NumPy, TensorFlow (Data Science/ML), Linux/Arch

Professional Experience

Backend Developer – African Population Health and Research Center

Remote | 2023 – Present

- Designed and developed scalable Django REST APIs for journal/article management, integrating platforms, countries, languages, and thematic areas.
- Implemented Celery task queues with Django + Redis for background jobs (file processing, notifications) and monitored tasks with Flower.
- Optimized Postgres queries (indexes, joins, schema design) for high-performance data retrieval.
- Built authentication and role-based access control, ensuring security across APIs.
- Developed React components with TypeScript for journal submission UI, integrated with Django APIs.

- Applied TDD practices: wrote unit/integration tests with Pytest and enforced code quality with Flake8.
- Containerized services with Docker Compose for local development and prepared apps for Kubernetes deployments.

MLOps Engineer – African Population Health and Research Center

Remote | 2020 – 2023

- Built machine learning pipelines for gesture recognition and time-series analysis using TensorFlow and Scikit-learn.
- Deployed ML models with Docker + Kubernetes, automating workflows with GitLab CI/CD.
- Integrated LangChain to experiment with LLM-powered query pipelines for structured outputs.
- Collaborated with cross-functional teams using Scrum (GitLab Boards) for sprint planning.

IT & Networking Intern – Ministry of Culture and Heritage

Nairobi, Kenya | 2019

- Configured and maintained small office networks, including Ethernet crimping, subnetting, TCP/IP, DNS, and DHCP.
- Provided support for hardware/software issues and optimized LAN/WAN connectivity.
- Gained foundational knowledge in IT infrastructure supporting backend deployment environments.

Education

BSc. Computer Science – Strathmore University

Graduated: 2022

Notable Projects

- Africa Research Connect Platform (Django + React + Celery + Docker) for connecting different researchers on African continent [Live Link](#)
- Journal Management System (Django + React + Celery + Docker) – Built a platform for African academic publishing, including API integrations, task queues, and analytics dashboards. [Live Link](#) | [Github Link](#)
- Chicken disease classifier (TensorFlow + Flask API) – Developed a web-based platform that classifies chicken fecal images to detect Coccidiosis using a deep learning model. Integrated Docker for containerization, DVC for data versioning, and deployed using an AWS CI/CD pipeline for automated updates and scalability. [Github Link](#)

Core Strengths

- Backend API Development (Django REST + Postgres/MySQL)
- Scalable Asynchronous Systems (Celery + Redis + Docker)
- Test-Driven Development (TDD, Flake8, CI/CD)
- Cross-functional Collaboration (Scrum, GitLab Boards)
- Cloud Deployments (GCP +AWS +Azure)
- LLM Enthusiast (LangChain/LangGraph projects)